

**WARNING:** Sorry, the genes E3.2.1.86A, algA, atpA, csgE, cyoB, fimA, fimY, gtrB, gtrC1, hyi, iagB, lpfD, mdtJ, miaA, mtlD, nfsB, oxlT, phoQ, prgl, psIB, rfbC, rfbD, rfbG, rfbN, rffA, rnr, sefA, sicP, sigE, siiC, siiD, siiF, smg, spaN, spiA, ssaD, ssaE, ssal, ssaK, ssaL, ssaT, ssaU, ssas, sseE, sseK1, sseL, ssrB, sthA\_05255, tomB, ugd, waaA, wbpW, xanB, yehD, znuB aren't in our dataset, which is focused on lead-operon promoters. Please check out the other [SalCom databases](#) for our other comprehensive transcriptomic data.

Reference:

nature microbiology

Resource  
**Profiling *Salmonella* transcriptional dynamics during macrophage infection using a comprehensive reporter library**

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[PDF \(15 MB\)](#)

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Welcome to SalComKinetics

SalcomKinetics is an interactive browser for the gene expression data presented in [Temporal profiling of \*Salmonella\* transcriptional dynamics during macrophage infection using a comprehensive reporter library](#). These data are collected from our GFP-reporter assay from 2,907 computationally identified promoter regions.

Instructions: Search by gene, locus or location (multiple entries separated by comma or space, use "\*" for wild-card).

\*To start: enter query or select **Example:** Top 100 genes in the early stage (1-4 h.p.i.) • Top 100 genes in the middle stage (5-9 h.p.i.) • Top 100 genes in the late stage (10-12 h.p.i.) • Top 100 genes in the escape stage (13-15 h.p.i.) • Enter-Doudoroff • Mn2+ regulation • SPI1 genes • SPI2 • SPI1 & 2 effectors • SPI3 • SPI5 • SPI6 • SPI11

algA, psIB, rfbA, wbpW, xanB, atpA, cls, cpxR, csgD, csgE, cspA, cyoB, dam, dksA, dnaK, dpaL, E3.2.1.86A, celF, fimA, fim

Query

Heat map for expression of 56 genes in SL1344

Macrophage Expression Levels

| query   | 01h | 02h | 03h | 04h | 05h  | 06h  | 07h  | 08h  | 09h  | 10h  | 11h  | 12h  | 13h  | 14h  | 15h  | Gene_Plate     | Gene        | Synonym | Ty2 Accession | LT2 Accession |
|---------|-----|-----|-----|-----|------|------|------|------|------|------|------|------|------|------|------|----------------|-------------|---------|---------------|---------------|
| 1 rfbA  | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | rfbA_P29       | rfbA        |         | T_RS03915     | STM209        |
| 2 cls   | -   | -   | -   | -   | -    | -    | -    | -    | -    | 0.7  | 0.9  | 0.8  | 0.7  | 0.6  |      | cls_P12        | cls         | clsA    | T_RS08410     | STM173        |
| 3 cpxR  | -   | -   | -   | -   | -    | 1.4  | 1.5  | 1.5  | 1.3  | 1.4  | 1.4  | 1.4  | 1.2  | 1.2  |      | cpxR_P17       | cpxR        |         | T_RS18080     | STM405        |
| 4 csgD  | -   | -   | -   | -   | -    | -    | -    | 1.1  | 1.6  | 1.7  | 2.7  | 2.9  | 2.5  | 2.6  |      | csgD_P11       | csgD        |         | T_RS09045     | STM114        |
| 5 cspA  | -   | -   | -   | -   | 4.7  | 2.9  | 2.4  | 2.2  | 1    | -    | -    | -    | -    | -    |      | P16B4_P16      | P16B4       |         | T_RS19690     | STM364        |
| 6 dam   | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    |      | dam_P30        | dam         |         | T_RS20450     | STM348        |
| 7 dksA  | -   | -   | -   | -   | 2.1  | 2.9  | 3.7  | 6.9  | 4.1  | 4.8  | 3.5  | 3.3  | 3.4  | 3    | 2.5  | dksA_P1        | dksA        |         | T_RS00975     | STM018        |
| 8 dnaK  | -   | -   | 4.8 | 4.5 | 4.9  | 10.1 | 13.7 | 23.4 | 15.1 | 12.1 | 15.4 | 12.8 | 13.2 | 12   | 13.2 | dnaK_P1        | dnaK        |         | T_RS00060     | STM001        |
| 9 dpaL  | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | dpaL_P27       | dpaL        |         | T_RS04880     | STM100        |
| 10 celF | -   | -   | 9.8 | 4.1 | 2.4  | -    | 1.5  | -    | -    | -    | -    | -    | -    | -    | -    | celF_P22       | celF        |         | T_RS06110     | STM131        |
| 11 fimC | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | fimC_P27       | fimC        |         | T_RS11790     | STM054        |
| 12 fimI | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | fimI_P2        | fimI        |         | T_RS11795     | STM054        |
| 13 fimZ | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | fimZ_P27       | fimZ        |         | T_RS11770     | STM054        |
| 14 galU | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | 1    | galU_P3        | galU        |         | T_RS08470     | STM175        |
| 15 gntK | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | gntK_P14       | gntK        |         | T_RS20235     | STM354        |
| 16 ssrA | -   | -   | -   | -   | 1.6  | 2    | 2.8  | 3.8  | 3.8  | 4.2  | 3.9  | 4.5  | 4.6  | 4    | 3.6  | ssrA_P11       | ssrA        |         | T_RS06435     | STM139        |
| 17 grxD | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | ydhD_P8        | ydhD        |         | T_RS06640     | STM143        |
| 18 guaA | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | guaA_P5        | guaA        |         | T_RS01750     | STM251        |
| 19 hilA | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | hilA_P19       | hilA        |         | T_RS14095     | STM287        |
| 20 hrpA | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | hrpA_P8        | hrpA        |         | T_RS07860     | STM164        |
| 21 invF | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | invF_P12       | invF        |         | T_RS14210     | STM289        |
| 22 invH | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | invH_P12       | invH        |         | T_RS14215     | STM290        |
| 23 lpp  | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | lpp_P4         | lpp         |         | T_RS06360     | STM137        |
| 24 matP | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | ycbG_P6        | ycbG        | matP    | T_RS09435     | STM106        |
| 25 ompR | -   | -   | -   | -   | -    | -    | -    | 1.1  | -    | -    | 0.7  | 0.6  | -    | 0.6  | -    | ompR_P15       | ompR        |         | T_RS20360     | STM350        |
| 26 pagD | -   | -   | -   | -   | 6.2  | 9.6  | 11.6 | 7.4  | 53.2 | 20.8 | 18.8 | 41.5 | 24.9 | 19.7 | 19.7 | pagD_P7        | pagD        |         | T_RS05705     | STM124        |
| 27 pagK | -   | -   | -   | 5.7 | 11.3 | 16.4 | 23.9 | 22.3 | 22.2 | 21.8 | 20.8 | 24.8 | 25.9 | 24.8 | 22.2 | pagK_P19       | pagK        |         |               | STM186        |
| 28 pagL | -   | -   | -   | -   | -    | -    | -    | -    | -    | -    | -    | -    | -    | 0.6  | 0.6  | SL1344_2221_P4 | SL1344_2221 |         | T_RS03140     | STM224        |
| 29 pagO | -   | -   | -   | -   | 3.9  | 6.8  | 9.9  | 13.7 | 14   | 17   | 10.7 | 14.4 | 10.6 | 11.5 | 10.7 | pagO_P8        | pagO        |         | T_RS05175     | STM186        |
| 30 phoP | -   | -   | -   | -   | -    | 15.2 | 31.7 | 45.1 | 78.8 | 88.3 | 77   | 77   | 74.3 | 63.5 | 55.6 | phoP_P6        | phoP        |         | T_RS08600     | STM123        |



Promoter activity dynamics throughout intracellular macrophage infection. During the early stage of infection (1-4 h.p.i.), activity was quantified as the background-subtracted GFP signal. Promoter activity during the middle, late, and escape stages (5-15 h.p.i.) was quantified as the background-subtracted GFP signal, normalized by background-subtracted dTomato signal to account for intracellular replication. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For more details about normalization, please see Methods [here](#).

| query           | 01h  | 02h  | 03h  | 04h  | 05h  | 06h  | 07h  | 08h  | 09h  | 10h  | 11h  | 12h   | 13h   | 14h   | 15h  | Gene_Plate      | Gene        | Synonym | Ty2 Accession | LT2 Accession |
|-----------------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|------|-----------------|-------------|---------|---------------|---------------|
| 31 <i>pipA</i>  | -    | -    | -    | -    | 1.8  | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | pipA_P27        | pipA        |         | T_RS09330     | STM108        |
| 32 <i>pipB</i>  | -    | -    | -    | -    | 3.7  | 8.6  | 13.9 | 26.6 | 39.9 | 65.9 | 78.5 | 101.6 | 103.6 | 104.7 | 92.6 | pipB2_P12       | pipB2       |         | T_RS13590     | STM278        |
| 33 <i>pipB</i>  | -    | -    | -    | -    | -    | 7.6  | 13.8 | 20.7 | 20.6 | 16.8 | 16.1 | 12.4  | 13.4  | 12    | 12.3 | pipB_P7         | pipB        |         | T_RS09325     | STM108        |
| 34 <i>pstC</i>  | -    | -    | -    | -    | 2.5  | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | pstC_P32        | pstC        |         | T_RS18650     | STM385        |
| 35 <i>pyrD</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | P21E10_P21      | P21E10      |         | T_RS09490     | STM105        |
| 36 <i>rbsK</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | rbsK_P16        | rbsK        |         | T_RS18950     | STM379        |
| 37 <i>rbsK</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | EMPTY_P16       | EMPTY       |         | NaN           | NaN           |
| 38 <i>rbsK</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | rbsK_P4         | rbsK        |         | T_RS18480     | STM388        |
| 39 <i>rbsK</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | orf408_P31      | orf408      | rbsK    | T_RS06865     | STM138        |
| 40 <i>rfc</i>   | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | rfc_P5          | rfc         |         | T_RS06195     | STM133        |
| 41 <i>serB</i>  | 26.9 | -    | 38.9 | 29.4 | 8.2  | 3.9  | 3.2  | 3.3  | 3.3  | 3.4  | 4    | 1.9   | 1.3   | 1.2   | 0.8  | SL1344_2174_P29 | SL1344_2174 |         | T_RS03335     | STM219        |
| 42 <i>sipD</i>  | 11.6 | 12.6 | 19.2 | 9.3  | 5.8  | 6.6  | 4.6  | 4.4  | 3.8  | 3.5  | 2.1  | 1.9   | 1.5   | 1.2   | 1.2  | sipD_P25        | sipD        |         | T_RS14130     | STM288        |
| 43 <i>slyA</i>  | -    | -    | -    | 5.9  | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | slyA_P2         | slyA        |         | T_RS06695     | STM144        |
| 44 <i>sodB</i>  | -    | -    | -    | -    | -    | -    | 1.2  | 1.2  | 1.6  | 1.7  | 1.8  | 2.4   | 2.5   | 2.4   | 2    | sodB_P11        | sodB        |         | T_RS06630     | STM143        |
| 45 <i>sopD2</i> | 7.7  | 5.9  | -    | -    | 2.7  | 4.2  | 5.3  | 6.7  | 10   | 10.7 | 11   | 7.6   | 7     | 7.6   | 7.7  | sopD2_P6        | sopD2       |         | T_RS09990     | STM097        |
| 46 <i>sopE</i>  | -    | -    | -    | -    | -    | 1.8  | 1.6  | 2    | -    | -    | -    | -     | -     | -     | -    | sopE_P13        | sopE        |         | T_RS21905     |               |
| 47 <i>sopE2</i> | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | sopE2_P8        | sopE2       |         | T_RS09265     | STM185        |
| 48 <i>spaM</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | P30G1_P30       | P30G1       |         | spaM          |               |
| 49 <i>ssaG</i>  | -    | -    | -    | -    | 10.2 | 19.2 | 35.8 | 47.9 | 40.5 | 40.6 | 35.8 | 33.3  | 29.9  | 31.4  | 29.7 | ssaG_P8         | ssaG        |         | T_RS06505     | STM140        |
| 50 <i>ssaM</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | ssaM_P2         | ssaM        |         | T_RS06540     | STM141        |
| 51 <i>ssaR</i>  | -    | -    | -    | 4.1  | 15.1 | 24.1 | 37.4 | 45.4 | 36.4 | 38.3 | 34.4 | 28.7  | 33.1  | 26.8  | 26.5 | ssaR_P2         | ssaR        |         | T_RS06570     | STM141        |
| 52 <i>sscB</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | sscB_P22        | sscB        |         | T_RS06490     | STM140        |
| 53 <i>sseB</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | P31E6_P31       | P31E6       |         | T_RS01620     | STM253        |
| 54 <i>sseJ</i>  | -    | -    | -    | -    | 10.1 | 22.7 | 33.4 | 51   | 54.8 | 49.7 | 38.2 | 44    | 42.8  | 42.8  | 34.2 | sseJ_P8         | sseJ        |         | T_RS08125     | STM163        |
| 55 <i>stfA</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | stfA_P27        | stfA        |         | T_RS00990     | STM019        |
| 56 <i>waaB</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | rfaB_P20        | rfaB        |         | T_RS19330     | STM371        |
| 57 <i>ychF</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | ychF_P6         | ychF        |         | T_RS05585     | STM178        |
| 58 <i>ynfM</i>  | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -    | -     | -     | -     | -    | ynfM_P8         | ynfM        |         | T_RS07160     | STM148        |
| 59 <i>znuC</i>  | -    | -    | -    | -    | 3.9  | 2.7  | 3.2  | 1.4  | -    | -    | -    | -     | -     | -     | -    | znuC_P5         | znuC        |         | T_RS05025     | STM189        |



Promoter activity dynamics throughout intracellular macrophage infection. During the early stage of infection (1-4 h.p.i.), activity was quantified as the background-subtracted GFP signal. Promoter activity during the middle, late, and escape stages (5-15 h.p.i.) was quantified as the background-subtracted GFP signal, normalized by background-subtracted dTomato signal to account for intracellular replication. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For more details about normalization, please see Methods [here](#).

In Vitro Expression Levels

| query         | 00h InSPI2 | 01h InSPI2 | 02h InSPI2 | 03h InSPI2 | 04h InSPI2 | 05h InSPI2 | 06h InSPI2 | 07h InSPI2 | 08h InSPI2 | 09h InSPI2 | 10h InSPI2 | 11h InSPI2 | 12h InSPI2 | 13h InSPI2 | 14h InSPI2 | 15h InSPI2 | 16h InSPI2 | 17h InSPI2 | 18h InSPI2 | 19h InSPI2 | 20h InSPI2 | 21h InSPI2 | 22h InSPI2 | 23h InSPI2 | 24h InSPI2 | 00h r    |
|---------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|----------|
| 1 <i>rfbA</i> | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -        |
| 2 <i>cls</i>  | 83         | -203       | -170       | -337       | -87        | -37        | 403        | 728        | 1,125      | 1,211      | 1,418      | 1,432      | 1,270      | 888        | 628        | 591        | 393        | 241        | 448        | 622        | 526        | 129        | 599        | 384        | 370        | 357      |
| 3 <i>cpxR</i> | 417        | 317        | 503        | 667        | 1,390      | 2,670      | 4,449      | 5,125      | 5,718      | 6,339      | 6,671      | 5,612      | 4,478      | 3,803      | 2,770      | 2,970      | 2,960      | 2,680      | 2,860      | 2,800      | 2,600      | 2,590      | 2,680      | 2,460      | 2,560      | 60       |
| 4 <i>csgD</i> | 30         | 73         | 180        | 673        | 1,213      | 2,467      | 4,753      | 6,530      | 7,676      | 7,129      | 8,029      | 7,661      | 6,751      | 5,153      | 4,110      | 4,060      | 4,140      | 3,690      | 3,770      | 3,940      | 3,550      | 3,660      | 3,550      | 3,620      | 3,620      | 40       |
| 5 <i>cspA</i> | 223        | 1,160      | 1,523      | 1,373      | 1,677      | 2,183      | 2,997      | 2,954      | 2,794      | 2,524      | 2,630      | 2,319      | 1,871      | 1,421      | 1,170      | 708        | 845        | 776        | 859        | 785        | 905        | 764        | 1,250      | 988        | 963        | -120 1,  |
| 6 <i>dam</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -        |
| 7 <i>dkSA</i> | 1,190      | 963        | 1,303      | 2,713      | 5,527      | 8,563      | 12,020     | 12,095     | 12,009     | 11,456     | 11,795     | 9,974      | 7,887      | 6,709      | 4,810      | 4,940      | 4,790      | 4,710      | 4,600      | 4,710      | 4,570      | 4,260      | 4,280      | 4,180      | 3,900      | 1,127 1, |
| 8 <i>dnaK</i> | 1,047      | 1,313      | 1,930      | 4,637      | 7,923      | 10,777     | 15,082     | 16,389     | 15,862     | 17,740     | 18,565     | 16,431     | 12,911     | 10,180     | 8,270      | 7,040      | 6,740      | 6,910      | 6,890      | 6,470      | 6,310      | 6,650      | 6,190      | 5,990      | 6,190      | 1,627 1, |



Promoter activity dynamics throughout growth in in vitro conditions (InSPI2, InSPI2 low Mg, NonSPI2, and 10% BHI). Activity was quantified as the background-subtracted GFP signal, normalized by background-subtracted OD600 to account for changes in cell density. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For visualization, one datapoint is shown per hour. For the dynamic data, please toggle to full resolution view to visualize all data points, or download the raw data. For more details about normalization please see Methods [here](#).

| query           | 00h InSPI2 | 01h InSPI2 | 02h InSPI2 | 03h InSPI2 | 04h InSPI2 | 05h InSPI2 | 06h InSPI2 | 07h InSPI2 | 08h InSPI2 | 09h InSPI2 | 10h InSPI2 | 11h InSPI2 | 12h InSPI2 | 13h InSPI2 | 14h InSPI2 | 15h InSPI2 | 16h InSPI2 | 17h InSPI2 | 18h InSPI2 | 19h InSPI2 | 20h InSPI2 | 21h InSPI2 | 22h InSPI2 | 23h InSPI2 | 24h InSPI2 | 00h InSPI2 |
|-----------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|------------|
| 9 <i>dpaL</i>   | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 10 <i>cellF</i> | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 11 <i>fimC</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 12 <i>fimI</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 13 <i>fimZ</i>  | -17        | 110        | 150        | 473        | 637        | 1,173      | 876        | 662        | 463        | 562        | 572        | -39        | 287        | -73        | -35        | 193        | 84         | 92         | 122        | -157       | 38         | 48         | -46        | -144       | -293       | -          |
| 14 <i>galU</i>  | -317       | -223       | -490       | -517       | -203       | -283       | 307        | 1,420      | 1,568      | 1,776      | 1,805      | 1,922      | 1,581      | 985        | 861        | 736        | 583        | 1,010      | 884        | 757        | 752        | 747        | 785        | 530        | 471        | 210        |
| 15 <i>gntK</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 16 <i>ssrA</i>  | 533        | 470        | 747        | 1,600      | 2,583      | 4,910      | 7,401      | 7,355      | 7,694      | 8,209      | 8,967      | 8,028      | 6,178      | 4,974      | 4,000      | 4,100      | 4,060      | 4,140      | 3,900      | 3,780      | 3,880      | 3,950      | 3,930      | 3,830      | 3,490      | 563        |
| 17 <i>grxD</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 18 <i>guaA</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 19 <i>hilA</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | 570        |
| 20 <i>hrpA</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 21 <i>invF</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 22 <i>invH</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 23 <i>lpp</i>   | 5,497      | 3,957      | 5,253      | 10,280     | 16,907     | 21,650     | 20,925     | 20,475     | 19,142     | 19,364     | 16,616     | 13,470     | 10,923     | 8,994      | 8,450      | 8,350      | 8,040      | 7,910      | 7,820      | 7,390      | 7,360      | 7,160      | 6,850      | 7,180      | 6,610      | 3,657      |
| 24 <i>matP</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 25 <i>ompR</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | 100        |
| 26 <i>pagD</i>  | -123       | -80        | -297       | -390       | -153       | -623       | 102        | 458        | 455        | 776        | 1,006      | 876        | 801        | 475        | 273        | 296        | 218        | 499        | 200        | 215        | 124        | 129        | 602        | 163        | 160        | 367        |
| 27 <i>pagK</i>  | -960       | -903       | -530       | 213        | 1,060      | 2,653      | 4,630      | 5,298      | 5,824      | 7,071      | 6,660      | 4,904      | 3,658      | 2,867      | 2,520      | 2,020      | 1,790      | 1,890      | 1,740      | 1,440      | 1,650      | 1,590      | 1,490      | 1,600      | 1,360      | 650        |
| 28 <i>pagL</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 29 <i>pagO</i>  | 340        | 277        | 320        | 420        | 737        | 1,497      | 1,854      | 2,185      | 2,424      | 2,096      | 1,969      | 2,224      | 1,565      | 1,123      | 1,240      | 788        | 554        | 807        | 610        | 696        | 576        | 658        | 498        | 724        | 626        | 17         |
| 30 <i>phoP</i>  | 23         | -250       | -130       | 253        | 1,760      | 3,033      | 5,823      | 7,559      | 7,519      | 6,940      | 7,505      | 7,198      | 5,936      | 4,326      | 3,560      | 3,240      | 2,960      | 3,100      | 3,200      | 3,290      | 3,260      | 2,890      | 3,140      | 2,900      | 2,880      | 317        |
| 31 <i>pipA</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 32 <i>pipB</i>  | -310       | -623       | -670       | -563       | -667       | -490       | 207        | 1,266      | 2,300      | 2,861      | 3,371      | 3,386      | 2,959      | 2,302      | 1,940      | 1,440      | 1,140      | 956        | 1,090      | 1,040      | 1,250      | 978        | 1,190      | 1,070      | 1,100      | 110        |
| 33 <i>pipB</i>  | -247       | -187       | -103       | -250       | 173        | 497        | 1,370      | 2,274      | 2,996      | 3,434      | 3,395      | 2,813      | 2,161      | 1,377      | 1,120      | 788        | 856        | 1,110      | 906        | 915        | 988        | 951        | 1,030      | 762        | 774        | -280       |
| 34 <i>pstC</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 35 <i>pyrD</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 36 <i>rbsK</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 37 <i>rbsK</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 38 <i>rbsK</i>  | 450        | 413        | 290        | 650        | 737        | 1,023      | 935        | 655        | 656        | 740        | 544        | 404        | 234        | 339        | 438        | 394        | 470        | 398        | 346        | 366        | 207        | 244        | 497        | 240        | 332        | -          |
| 39 <i>rbsK</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 40 <i>rfc</i>   | 370        | 293        | 110        | 297        | 247        | 33         | 577        | 499        | 805        | 765        | 693        | 744        | 901        | 826        | 590        | 264        | 594        | 657        | 230        | 281        | 367        | 529        | 370        | 330        | 150        | -          |
| 41 <i>serB</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 42 <i>sipD</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 43 <i>slyA</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 44 <i>sodB</i>  | 167        | 47         | 83         | 83         | 163        | 1,090      | 2,443      | 4,443      | 4,719      | 4,753      | 5,669      | 5,055      | 4,375      | 3,314      | 2,580      | 2,520      | 2,440      | 2,420      | 2,520      | 2,740      | 2,370      | 2,500      | 2,490      | 2,640      | 2,140      | 30         |
| 45 <i>sopD2</i> | -630       | -767       | -690       | -783       | -620       | -567       | 497        | 997        | 2,272      | 3,022      | 3,945      | 3,426      | 3,065      | 2,390      | 1,960      | 1,690      | 1,460      | 1,560      | 1,730      | 1,490      | 1,790      | 1,550      | 1,850      | 1,440      | 1,480      | 160        |
| 46 <i>sopE</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | 923        |
| 47 <i>sopE2</i> | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 48 <i>spaM</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 49 <i>ssaG</i>  | 190        | 187        | 373        | 460        | 1,097      | 2,050      | 3,785      | 5,971      | 7,562      | 8,849      | 8,553      | 7,864      | 5,844      | 4,452      | 3,330      | 3,100      | 2,900      | 3,070      | 2,890      | 2,790      | 2,870      | 2,710      | 2,400      | 2,790      | 2,580      | 447        |
| 50 <i>ssaM</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 51 <i>ssaR</i>  | -97        | -230       | 43         | 107        | 560        | 1,577      | 4,507      | 5,774      | 6,967      | 7,815      | 7,464      | 6,477      | 5,264      | 4,132      | 3,220      | 3,220      | 3,090      | 3,050      | 3,300      | 3,250      | 3,150      | 3,470      | 3,200      | 3,000      | 3,170      | -177       |
| 52 <i>sscB</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 53 <i>sseB</i>  | 427        | 100        | 270        | 430        | 747        | 670        | 543        | 452        | 222        | 38         | 344        | -78        | 150        | 200        | 200        | 288        | 291        | 299        | 338        | 356        | 168        | 182        | 416        | 142        | 147        | -          |
| 54 <i>sseJ</i>  | -140       | 30         | 153        | 147        | 353        | 1,203      | 3,517      | 6,409      | 9,000      | 11,198     | 11,935     | 11,903     | 9,871      | 7,460      | 5,790      | 5,700      | 5,480      | 5,250      | 5,280      | 5,240      | 5,220      | 5,420      | 5,270      | 5,290      | 5,100      | -3         |
| 55 <i>stfA</i>  | 30         | 157        | 163        | 313        | 837        | 990        | 369        | 557        | 561        | 374        | 386        | 12         | 221        | 166        | 81         | 281        | 176        | 197        | -29        | -175       | 126        | 45         | 269        | -62        | -153       | -          |
| 56 <i>waaB</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 57 <i>ychF</i>  | -313       | -400       | -213       | -363       | 207        | 713        | 1,592      | 1,474      | 1,545      | 1,369      | 1,828      | 1,692      | 1,248      | 1,221      | 703        | 577        | 674        | 528        | 586        | 443        | 623        | 386        | 720        | 481        | 764        | 133        |
| 58 <i>ynfM</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |
| 59 <i>znuC</i>  | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          | -          |



Promoter activity dynamics throughout growth in in vitro conditions (InSPI2, InSPI2 low Mg, NonSPI2, and 10% BHI). Activity was quantified as the background-subtracted GFP signal, normalized by background-subtracted OD600 to account for changes in cell density. Gray boxes indicate that the signal is "OFF," meaning the promoter activity is below the dynamic background threshold. For visualization, one datapoint is shown per hour. For the dynamic data, please toggle to full resolution view to visualize all data points, or download the raw data. For more details about normalization please see Methods [here](#).